

Nested Class

Nested classes are of 4 types:

1. Static Nested Class.

```
public class NestedClass {
    public static void main(String[] args) {
        Outer outer = new Outer(); // to access outer class
        Outer.Inner inner = new Outer.Inner();// to access the nested inner class.
        inner.fun();// me inner nested class ki function ko call kr dega.
    }
}

class Outer{
    static class Inner{ //ye nested class static ha to isko access krne k lia hum outer class ke object ki zarurat
nahi ha.
        void fun(){
            System.out.println("Hello");
        }
    }
}
```

- Static inner class can't access the non-static members of the outer class directly.
- To get access we have to take parameters in methods or constructors of inner class.
- Inner class behaves same as normal class.
- Inner class ka nam uski outer class ke sath hi juda hoga, example→Outer.Inner.
- It can act as helper class for outer class.
- It is used in builder design pattern.
- If you want to have static methods inside a nested class.
- Request / Response DTO.

2. Inner Nested Class: The class which is not static-nested class.

```
public class InnerNestedClass {
    public static void main(String[] args) {
        Outer outer = new Outer(); // outer class ka object simple hi bnega.
        Outer.Inner inner = outer.new Inner();// yaha pr nested class k lia
        // "new" se phle hume "outer" referance likhna zaruri ha.
        inner.fun();
    }
}

class Outer{
    int x =10;
    class Inner{ // is class me static nahi ha to ye outer class ke object pr depend krte ha.
        int x = 20;
        void fun(){
            System.out.println(x);
            System.out.println(Outer.this.x); //to call outer class variable.
        }
    }
}
```

- It can access Static and non-Static members of the outer class.

3. Local Nested Class: koi bhi asi class jo hum kisi code block ke ander bna dege, Example→ if, for, while, switch, function, method, constructor, static block, etc.

```
public class LocalNestedClass {
    public static void main(String[] args) {
        Outer outer = new Outer();
        outer.greet();
    }
}

class Outer {
    private int x =4;
    void greet(){// is method ke ander nested local class ha.
```

```

int y = 5; // me is local variable ko is local class me use to kr skta hu pr,
//y++; // me is "y" variable ko modify nahi kr skta. {EFFECTIVE FINAL VARIABLE} rule se.
class Local { // is class ko local nested class khte ha.
    void sayHello() {
        System.out.println(y);
    }
}
Local local = new Local(); // local class ka object hmesha resoective orgainization ke ander hi rhega.
local.sayHello();
}
}

```

Bhot kum use case ha local nested class ka.

- It uses **effective final variable** rule:

4. Anonymous Nested class: A class without Name is called Anonymous class.

```

public class AnonymusNestedClass {
    public static void main(String[] args) {
        Person p1 = new Person(){ // this is a anonymous class with no name.
            @Override
            void introduce(){
                greet(); // muj greet yhi call krna hoga .
                System.out.println("Hii i am Priyansh");
            }
            void greet(){
                System.out.println("Hello"); // me bahar greet call nhi kr skta.
            }
        };
        p1.introduce();
    }
}

class Person {
    void introduce(){
        System.out.println("HI I am a guest");
    }
}

```

- This class automatically extends the class.
- This class can't call constructor, kuki usme object name nhi ha.

Benefits of Nested classes:

1. Logical grouping.
2. Get better access of Outer class, example→private keyword, therefore it does not requires any getters/setters of Outer class.